CRIMEGRAPH

TECHNICAL SPECIFICATION & ARCHITECTURE BREAKDOWN (v1.1.0)

CrimeGraph is a standalone, mobile-first intelligence mapping platform designed for law enforcement field operatives and intelligence analysts. It is engineered with a strict offline-first, zero-trust architecture to ensure data sovereignty and compliance with UK policing digital strategies.

FRAMEWORK	`React 18 + TypeScript`
NATIVE BRIDGE	`Capacitor 6 (Android API Level 34+)`
STATE MANAGEMENT	`Zustand`
VISUALISATION ENGINE	`Cytoscape.js (WebGL Accelerated)`
DATABASE	`Capacitor SQLite (Device-Local)`
CRYPTOGRAPHY	`Web Crypto API (AES-GCM 256-bit)`

1. ARCHITECTURE & DATA SOVEREIGNTY

The application operates entirely independently of external servers or cloud infrastructure. All data parsing, relationship mapping, and state synchronisation occur directly on the device's CPU/GPU.

- **Zero-Cloud Topology:** CrimeGraph contains no outbound network APIs, telemetry, or remote analytics. Data cannot be intercepted in transit because data never traverses a network during standard operation.
- **Local SQLite Storage:** The platform utilises a heavily structured, relational SQLite database deployed directly to the application's secure sandbox on the Android filesystem.

2. SECURITY & ENCRYPTION PROTOCOLS

Authentication (Biometric Wrapper)

Access is gated via `@capacitor-community/biometric-auth`. The application verifies the operator's identity against the hardware-backed Android Keystore (Fingerprint/FaceID) prior to initialising the React lifecycle. Idle timeouts force re-authentication.

Intelligence Export Cryptography

When analysts share case files, the raw JSON data is encrypted using **AES-GCM (Advanced Encryption Standard - Galois/Counter Mode)**. GCM provides both confidentiality and data authenticity, ensuring the intelligence package has not been tampered with in transit. The derivation relies on a user-provided, one-time symmetric key.

Sanitisation (The Kill Switch)

The system features a catastrophic data destruction protocol. When triggered, the application executes rapid `DELETE FROM` commands across all local SQLite tables (cases, nodes, edges, notes, audit_logs), rendering the intelligence forensically irrecoverable.

3. DATA MODELLING (POLE FRAMEWORK)

The database schema is strictly aligned with the UK policing POLE methodology.

- **Nodes Table:** Stores entities (Person, Object, Location, Event). Includes a dynamic JSON stringified `attributes` column to house infinite custom metadata fields (e.g., URNs, vehicle registrations).
- **Edges Table:** Stores unidirectional and bidirectional vectors defining relationships between nodes.
- **Notes Table (v1.1):** Stores narrative intelligence logs, featuring an array of tagged node IDs for structural relational mapping.

4. AUDIT & DISCLOSURE (CPIA COMPLIANCE)

To adhere to the Criminal Procedure and Investigations Act 1996, CrimeGraph maintains a background, read-only `audit_logs` table. Every CRUD (Create, Read, Update, Delete) operation invokes the `logAudit()` function, permanently stamping the action, timestamp, and target entity ID into the local ledger to maintain a robust chain of evidence.